

ICT171 Assignment 3 – Cloud Server Project and Video Explainer

Personal Portfolio Website Hosted on Microsoft Azure

Student Name: Clinton Narankaik

Student Number: 34874536

GitHub Repository: <https://github.com/narankaik/ICT171-Cloud-Server-Project.git>

Public IP Address: <http://20.70.155.162/>

DNS URL: <http://clintonportfolio2026.australiaeast.cloudapp.azure.com>

Website URL

<https://clintonportfolio2026.australiaeast.cloudapp.azure.com>

Video Explainer:

[ICT171 video recording.mp4](#)

1. Introduction

Cloud computing has become a fundamental component of modern information technology infrastructure, providing organisations and individuals with scalable, cost-effective, and highly available computing resources. Infrastructure as a Service (IaaS) enables users to provision and manage virtual servers without the need to purchase or maintain physical hardware.

The purpose of this project was to design, deploy, and document a cloud-hosted personal portfolio website using Microsoft Azure. The project required the creation of a Linux virtual machine, configuration of networking and security settings, installation of a web server, deployment of a website, implementation of Domain Name System (DNS) services, and configuration of Secure Sockets Layer/Transport Layer Security (SSL/TLS) encryption.

The project was fully documented using GitHub to demonstrate version control practices and to provide sufficient information for another user to recreate the server environment. A Bash monitoring script was also developed to automate basic server health checks.

The successful completion of this project demonstrates practical skills in cloud infrastructure deployment, Linux server administration, web hosting, network security, scripting, and technical documentation.

2. Azure Virtual Machine Creation

2.1 Objective

The first stage of the project involved creating a cloud-based virtual machine within Microsoft Azure. The virtual machine serves as the foundation for hosting the portfolio website and supporting all server-side services.

2.2 Virtual Machine Configuration

The virtual machine was deployed using Microsoft Azure's Virtual Machine service. Ubuntu Server 24.04 LTS was selected due to its stability, security, and widespread use within server environments.

Configuration Summary

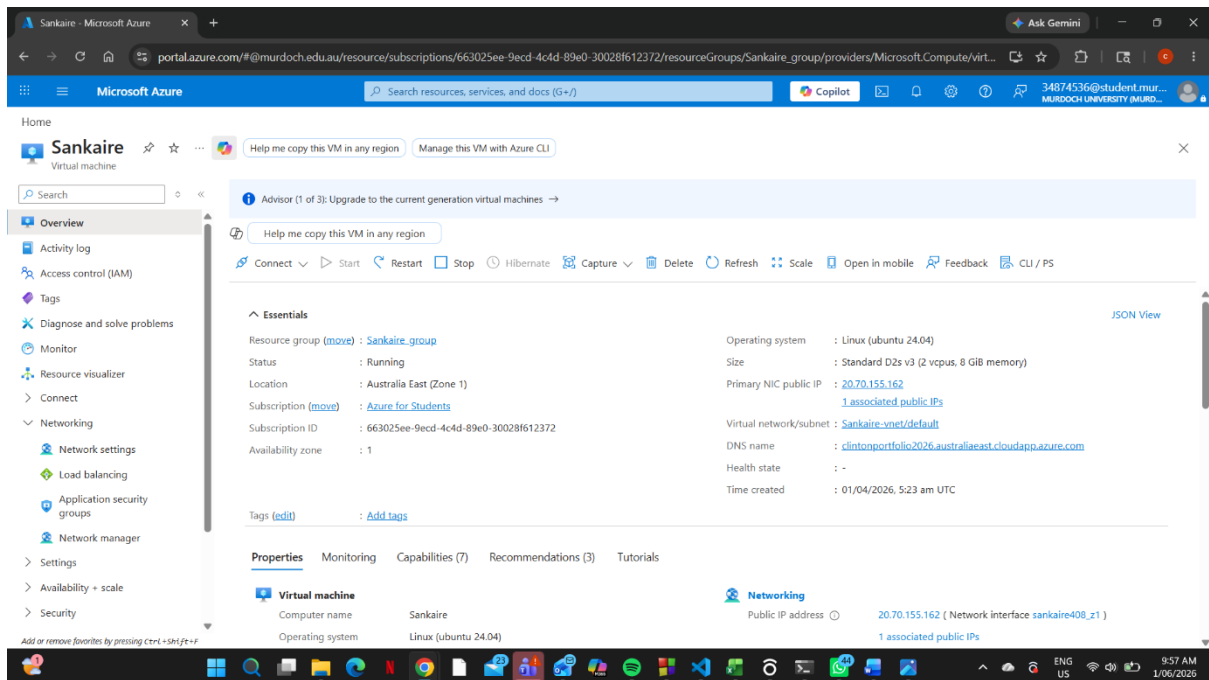
Parameter	Value
Cloud Provider	Microsoft Azure
Operating System	Ubuntu Server 24.04 LTS
Region	Australia East
Authentication	SSH Public Key
VM Size	B1s
Public IP Address	Dynamic
Storage Type	Standard SSD

2.3 Deployment Process

The deployment process involved creating a Resource Group, selecting the operating system image, configuring authentication credentials, and assigning a public IP address. Azure automatically provisioned the virtual machine and associated networking resources.

Figure 1

Azure Virtual Machine Overview



The successful deployment of the virtual machine provided a secure and scalable platform for hosting the portfolio website.

3. Network Security Group Configuration

3.1 Objective

Network Security Groups (NSGs) were configured to control inbound traffic to the virtual machine. These rules ensure that only authorised traffic can access the server.

3.2 Configured Rules

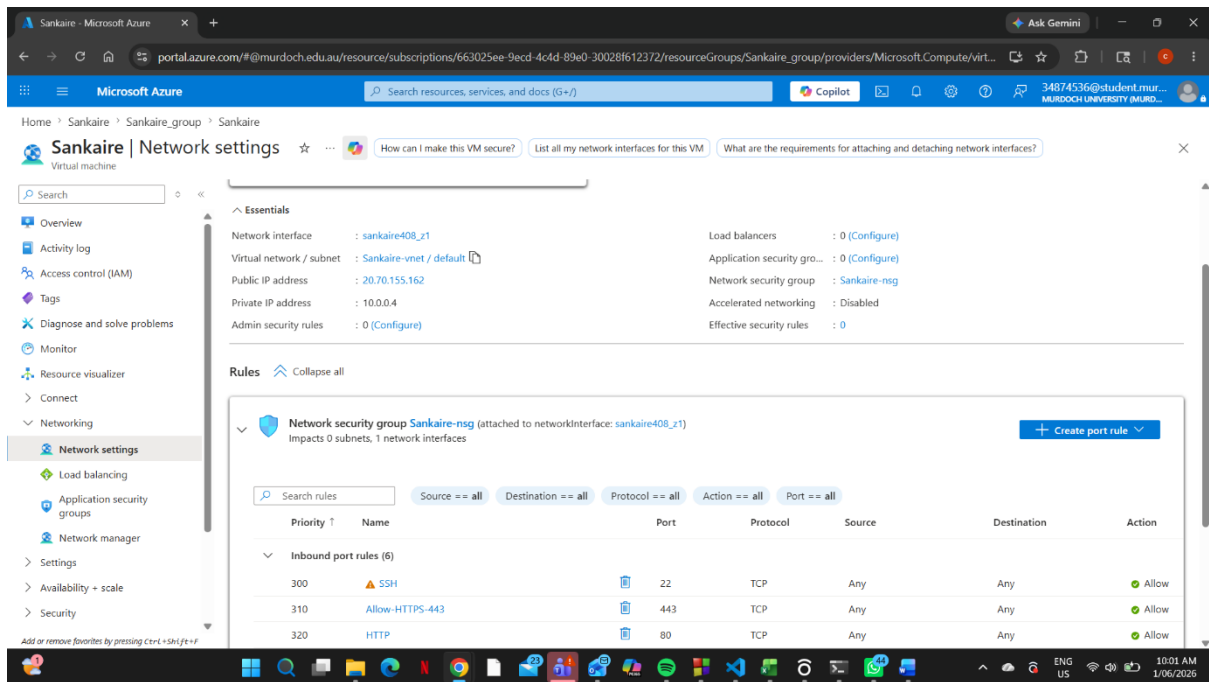
The following inbound rules were configured:

Port Protocol Purpose

22	TCP	Secure Shell (SSH)
80	TCP	Hypertext Transfer Protocol (HTTP)
443	TCP	Hypertext Transfer Protocol Secure (HTTPS)

Figure 2

Network Security Group



By restricting access to only the required ports, the attack surface of the server was significantly reduced.

4. SSH Access Setup

4.1 Objective

SSH was used to remotely administer the Linux server through a secure encrypted connection.

4.2 Connection Method

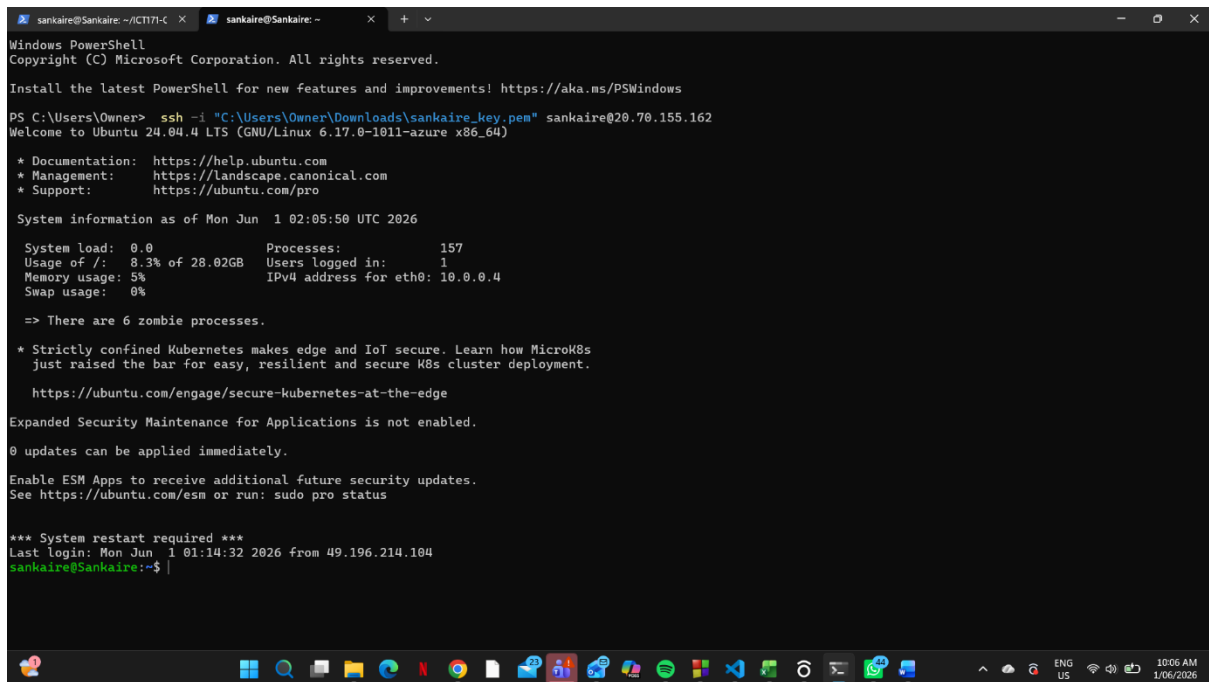
The following command was used to establish a secure SSH session:

```
ssh -i sankaire_key.pem
azureuser@clintonportfolio2026.australiaeast.cloudapp.azure.com
```

Successful authentication confirmed that the SSH key pair had been configured correctly.

Figure 3

Successful SSH Connection



```
sankaire@Sankaire: ~/ICT171-C x sankaire@Sankaire: - x + v
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Owner> ssh -i "C:\Users\Owner\Downloads\sankaire_key.pem" sankaire@20.70.155.162
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1011-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Jun  1 02:05:50 UTC 2026

System load:  0.0          Processes:    157
Usage of /:   8.3% of 28.0GB Users logged in: 1
Memory usage: 5%          IPv4 address for eth0: 10.0.0.4
Swap usage:  0%

==> There are 6 zombie processes.

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***
last login: Mon Jun  1 01:14:32 2026 from 49.196.214.104
sankaire@Sankaire:~$
```

SSH provided secure command-line access for server administration throughout the project.

5. Nginx Installation and Configuration

5.1 Objective

Nginx was selected as the web server software responsible for serving website content to users.

5.2 Installation

The following commands were used to install Nginx:

```
sudo apt update
```

```
sudo apt install nginx -y
```

5.3 Verification

The service status was verified using:

```
sudo systemctl status nginx
```

The service was confirmed to be active and configured to start automatically during system boot.

Figure 4

Nginx Service Status

```
sankaire@Sankaire:~$ sudo apt update
sudo apt upgrade -y
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5764 B]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble-security/main amd64 Components [42.5 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Fetched 1076 kB in 1s (1317 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
sankaire@Sankaire:~$ sudo apt install nginx -y
sudo systemctl status nginx
sudo systemctl enable nginx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nginx is already the newest version (1.24.0-2ubuntu7.8).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
* nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset>
   Active: active (running) since Sun 2026-05-31 04:28:06 UTC; 19h ago
     Docs: man:nginx(8)
   Process: 16466 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master>
   Process: 16468 ExecStart=/usr/sbin/nginx -g daemon on; master_process o>
   Main PID: 16469 (nginx)
     Tasks: 3 (limit: 9445)
    Memory: 5.7M (peak: 6.3M)
```

Nginx provides excellent performance, low resource consumption, and strong support for modern web applications.

6. Website Deployment

6.1 Objective

The purpose of this stage was to deploy a personal portfolio website onto the web server.

6.2 Deployment Process

Website files were uploaded to the default Nginx web root directory:

```
/var/www/html
```

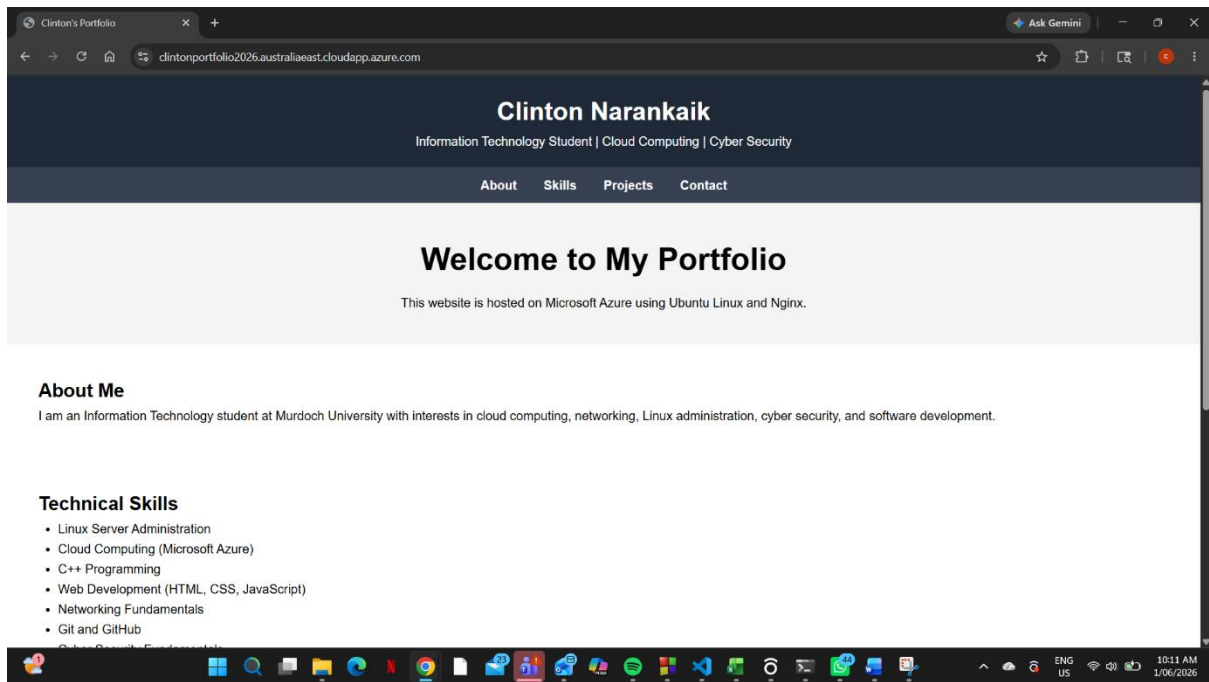
Files were copied using:

```
sudo cp index.html /var/www/html/
```

The website was then tested by accessing the DNS address through a web browser.

Figure 5

Portfolio Website Homepage



Successful deployment confirmed that the web server was correctly serving website content.

7. DNS Configuration

7.1 Objective

A DNS name was configured to provide a user-friendly address for accessing the website.

7.2 DNS Entry

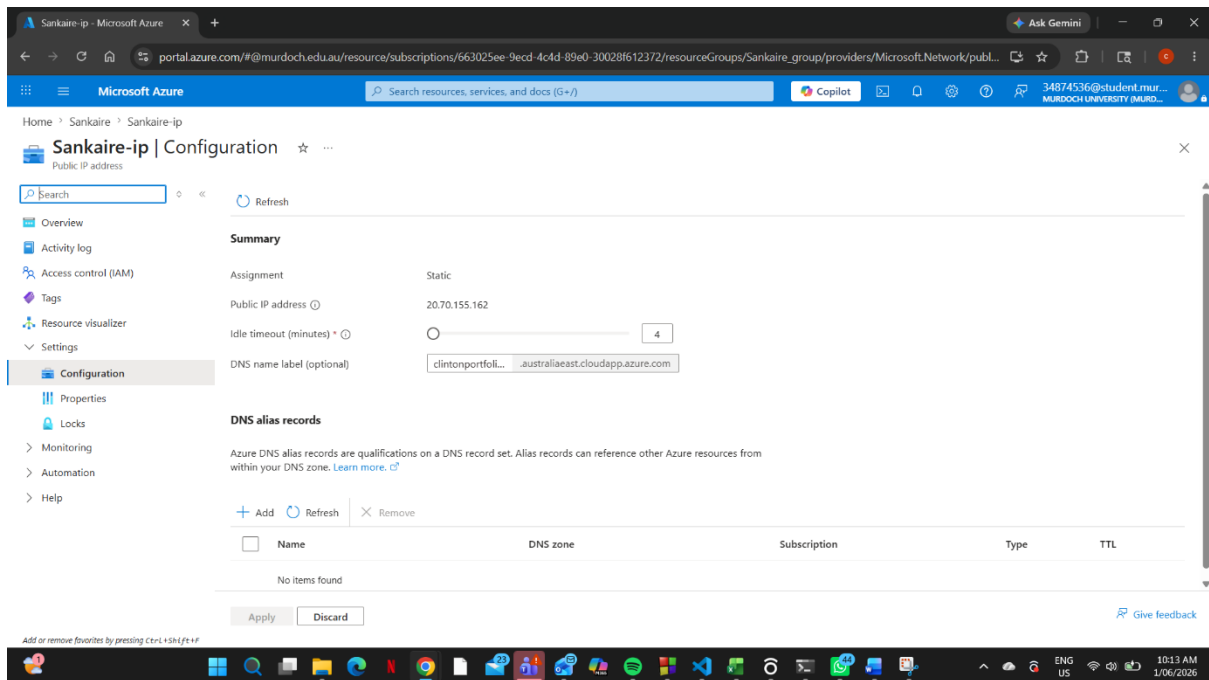
The following DNS name was assigned:

clintonportfolio2026.australiaeast.cloudapp.azure.com

The DNS entry was configured through Azure Public IP settings.

Figure 6

Azure DNS Configuration



The DNS name allows users to access the website without needing to remember a numerical IP address.

8. SSL/TLS Configuration

8.1 Objective

SSL/TLS encryption was implemented to secure communication between clients and the web server.

8.2 Certificate Installation

Certbot and Let's Encrypt were used to obtain a trusted SSL certificate.

Installation:

```
sudo apt install certbot python3-certbot-nginx -y
```

Certificate generation:

```
sudo certbot --nginx -d clintonportfolio2026.australiaeast.cloudapp.azure.com
```



```
sankaire@Sankaire: ~/CT171-C x sankaire@Sankaire: - x + v
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***
Last login: Mon Jun 1 01:14:32 2026 from 49.196.214.104
sankaire@Sankaire:~$ ls -la /var/www/html
total 12
drwxr-xr-x 2 www-data www-data 4096 Jun 1 01:16 .
drwxr-xr-x 3 root root 4096 Apr 7 12:45 ..
-rw-r--r-- 1 www-data www-data 2485 Jun 1 01:22 index.html
sankaire@Sankaire:~$ sudo certbot certificates
usage:
  certbot [SUBCOMMAND] [options] [-d DOMAIN] [-d DOMAIN] ...

Certbot can obtain and install HTTPS/TLS/SSL certificates. By default,
it will attempt to use a webserver both for obtaining and installing the
certificate.
certbot: error: unrecognized arguments: certificates
sankaire@Sankaire:~$ sudo certbot certificates
Saving debug log to /var/log/letsencrypt/letsencrypt.log

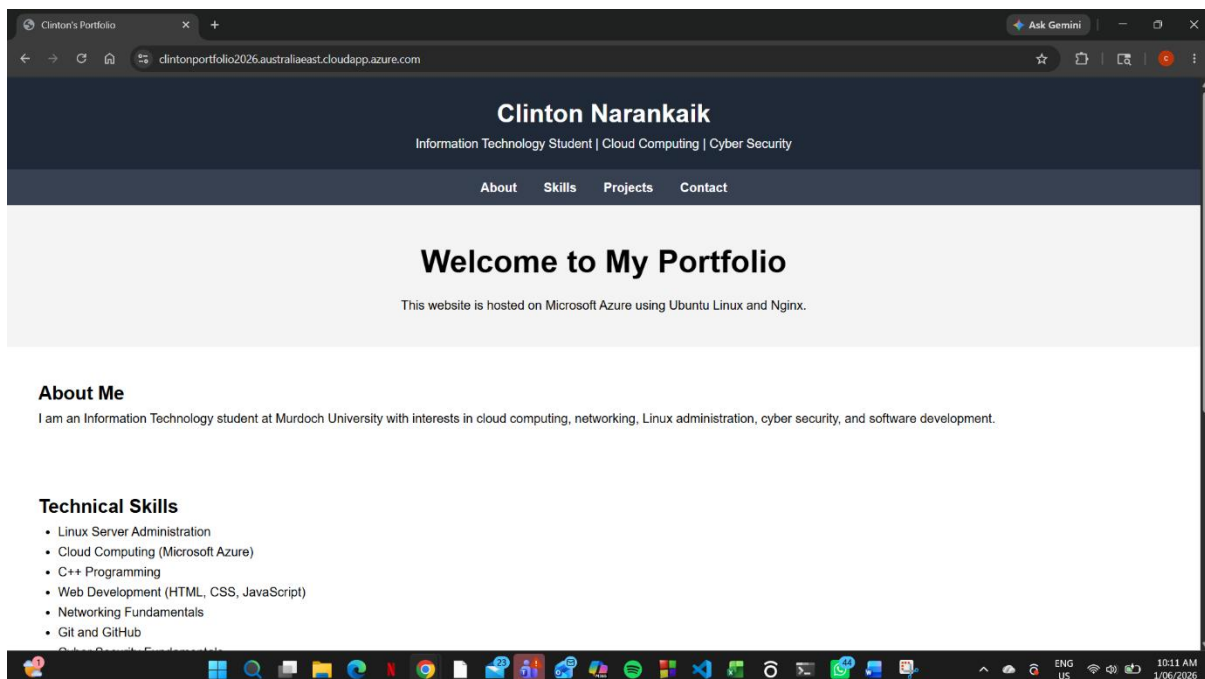
-----
Found the following certs:
Certificate Name: clintonportfolio2026.australiaeast.cloudapp.azure.com
Serial Number: 5551580e601f98f11a344a5d81838ce398c
Key Type: ECDSA
Domains: clintonportfolio2026.australiaeast.cloudapp.azure.com
Expiry Date: 2026-08-29 03:17:03+00:00 (VALID: 89 days)
Certificate Path: /etc/letsencrypt/live/clintonportfolio2026.australiaeast.cloudapp.azure.com/fullchain.pem
Private Key Path: /etc/letsencrypt/live/clintonportfolio2026.australiaeast.cloudapp.azure.com/privkey.pem
-----
sankaire@Sankaire:~$ |
```

8.3 Verification

The website was successfully accessed using HTTPS.

Figure 7

HTTPS Secured Website



Encryption ensures that user traffic is protected against interception and tampering.

9. Monitoring Script

9.1 Objective

A Bash script was developed to verify the operational status of the Nginx web server.

9.2 Script Code

```
#!/bin/bash
```

```
echo "Checking Nginx web server status..."
```

```
if systemctl is-active --quiet nginx
```

```
then
```

```
    echo "Nginx is running successfully."
```

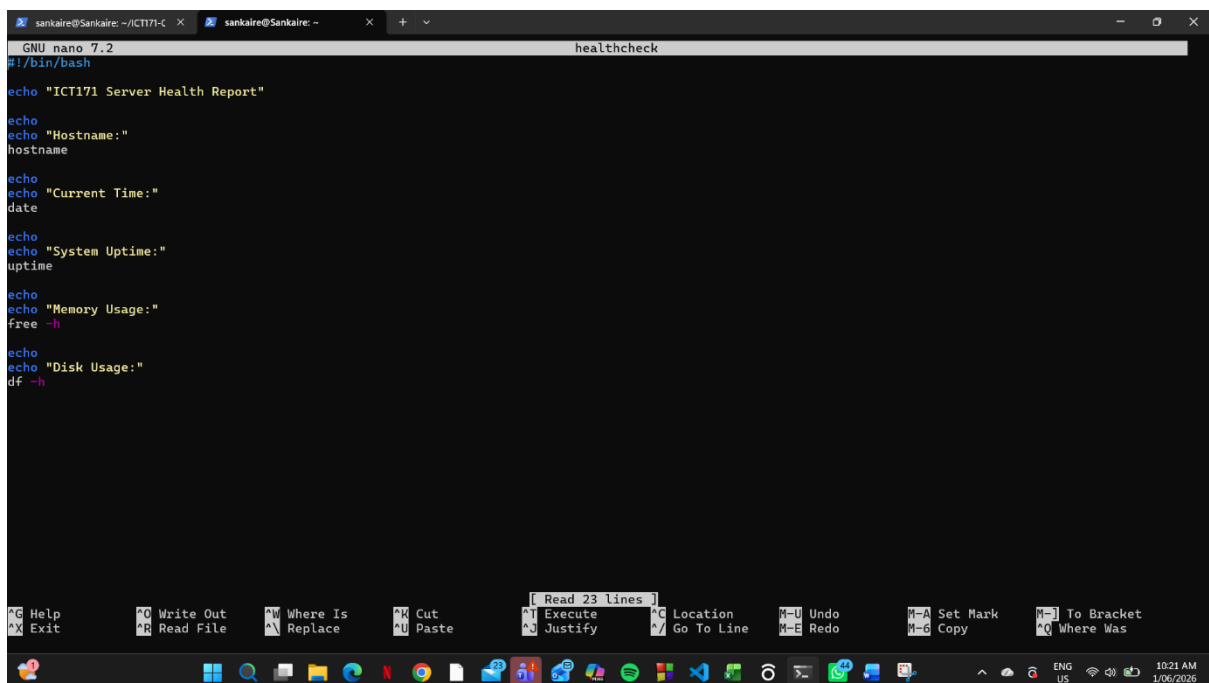
```
else
```

```
    echo "Nginx is not running."
```

```
fi
```

```
echo "Health check completed."
```

Monitoring script Source code

A screenshot of a terminal window with a dark background. The window title is "healthcheck". The terminal shows the source code of a Bash script being edited in the nano 7.2 editor. The script starts with a shebang line, followed by an echo statement for a health report, and then several echo statements for system information like hostname, date, uptime, memory usage (free -h), and disk usage (df -h). The bottom of the window shows the nano editor's menu bar with various shortcuts like Help, Exit, Write Out, Read File, Where Is, Replace, Cut, Paste, Execute, Justify, Location, Go To Line, Undo, Redo, Set Mark, Copy, To Bracket, and Where Was. The system tray at the very bottom shows the date and time as 10:21 AM on 1/06/2025.

9.3 Purpose

The script automates server health verification and assists administrators in identifying service outages quickly.

Figure 8

Monitoring Script Output

```
sankaire@sankaire:~$ ./healthcheck.sh
ICT171 Server Health Report

Hostname:
Sankaire

Current Time:
Mon Jun  1 02:23:50 UTC 2026

System Uptime:
02:23:50 up 1 day, 29 min,  5 users,  load average: 0.01, 0.00, 0.00

Memory Usage:
total      used      free     shared  buff/cache   available
Mem:    7.7Gi    629Mi    5.9Gi     4.1Mi     1.5Gi       7.1Gi
Swap:    0B         0B         0B

Disk Usage:
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        20G   2.4G   16G   12% /
tmpfs            3.9G   0     3.9G   0% /dev/shm
tmpfs            1.6G   1.1M   1.6G   1% /run
tmpfs            5.0M   0     5.0M   0% /run/lock
efivarfs         128K   38K   86K   31% /sys/firmware/efi/efivars
/dev/sda16       881M  117M   703M  15% /boot
/dev/sda15       105M   6.2M   99M   6% /boot/efi
/dev/sdb1        16G   28K   15G   1% /mnt
tmpfs            790M   12K   790M   1% /run/user/1000
```

10. Challenges Encountered and Solutions

Several challenges were encountered during the project implementation.

Challenge 1: SSH Key Permission Error

SSH initially rejected the private key due to incorrect permissions.

Solution: Corrected the permissions and re-established the connection.

Challenge 2: HTTP Port Accessibility

The website was initially inaccessible through a browser.

Solution: Added an inbound rule allowing HTTP traffic on port 80.

Challenge 3: SSL Configuration Error

An Nginx configuration error prevented successful SSL deployment.

Solution: Modified the Nginx configuration and redeployed the certificate successfully.

Challenge 4: GitHub Synchronisation

Commits were created locally but not visible in GitHub.

Solution: Pushed local commits to the remote repository using Git.

These challenges provided valuable troubleshooting experience and reinforced the importance of systematic problem-solving.

11. Conclusion

This project successfully demonstrated the deployment and administration of a cloud-based Linux web server using Microsoft Azure Infrastructure as a Service. The completed solution includes a publicly accessible website, DNS configuration, SSL/TLS encryption, remote SSH administration, GitHub documentation, and a custom monitoring script.

The project enhanced practical skills in cloud computing, Linux administration, networking, web hosting, scripting, and technical communication. These skills are directly relevant to modern industry practices and provide a strong foundation for future server and cloud-based projects.

12. References

Canonical. (2026). Ubuntu Server Documentation.

Certbot. (2026). Certbot Documentation.

GitHub. (2026). GitHub Documentation.

Microsoft. (2026). Azure Virtual Machines Documentation.

Nginx. (2026). Nginx Documentation.

Let's Encrypt. (2026). Let's Encrypt Documentation.

Appendices

Appendix A – Azure VM Screenshots

Appendix B – Network Security Group Screenshots

Appendix C – SSH Terminal Screenshots

Appendix D – Nginx Configuration Screenshots

Appendix E – Website Deployment Screenshots

Appendix F – DNS Configuration Screenshots

Appendix G – SSL/TLS Screenshots

Appendix H – Monitoring Script Screenshots